

Task-1: Certain experimental values of x and y are given below:

(0, -1), (2, 5), (5, 12), (7, 20)

If the straight-line $Y=a_0+a_1x$ is fitted to the above data, find the approximate values of a_0 & a_1

Code (Python):

```
from prettytable import PrettyTable
value_x=[0,2,5,7]
value_y=[-1,5,12,20]
def straight_line():
    table=PrettyTable()
    table.field_names=["x","y","x*y","x^2"]
    m=len(value_x)
    x_2=0
    sum_x=0
    xy=0
    sum_y=0
    for i in range(m):
        x,y=value_x[i],value_y[i]
        x_2+=x**2
        xy+=x*y
        sum_x+=x
        sum_y+=y
        table.add_row([x,y,x*y,x**2])
    table.add_row([f"Σx={sum_x}",
                   f"Σy={sum_y}",f"Σxy={xy}",
                   f"Σx^2={x_2}"])
    a1=(m*xy-sum_x*sum_y)/(m*x_2-sum_x**2)
    a0=(sum_y-sum_x*a1)/m
    print(table)
    print(f'a0={a0:.5f}\na1={a1:.5f}')

straight_line()
```

Output:

x	y	x*y	x^2
0	-1	0	0
2	5	10	4
5	12	60	25
7	20	140	49
Σx=14	Σy=36	Σxy=210	Σx^2=78

a0=-1.13793
a1=2.89655

Task-2: Fit a polynomial of the second degree to the data points (x, y) given by (0, 1), (1, 6) and (2, 17)

Code (Python):

```
import numpy as np
from prettytable import PrettyTable
value_x=[0,1,2]
value_y=[1,6,17]
def parabolic():
    table=PrettyTable()
    table.field_names=["x","y","x^2","x^3","x^4","x*y","x^2*y"]
    m=len(value_x)
    x_2=0
    x_3=0
    x_4=0
    x2y=0
    sum_x=0
    xy=0
    sum_y=0
    for i in range(m):
        x,y=value_x[i],value_y[i]
        x_2+=x**2
        xy+=x*y
        sum_x+=x
        sum_y+=y
        x_3+=x**3
        x_4+=x**4
        x2y+=(x**2)*y
        table.add_row([x,y,x**2,x**3,x**4,x*y,(x**2)*y])
    table.add_row([f"Σx={sum_x}",f"Σy={sum_y}",
                    f"Σx^2={x_2}",f"Σx^3={x_3}",
                    f"Σx^4={x_4}",f"Σxy={xy}",
                    f"Σx^2*y={x2y}"])
    A=np.array([[m,sum_x,x_2],
                [sum_x,x_2,x_3],
                [x_2,x_3,x_4]])
    B=np.array([sum_y,xy,x2y])
    X=np.linalg.solve(A,B)
    print(table)
    print(f'a0={X[0]:.2f}\na1={X[1]:.2f}\na2={X[2]:.2f}')

parabolic()
```

Output:

x	y	x^2	x^3	x^4	$x*y$	x^2*y
0	1	0	0	0	0	0
1	6	1	1	1	6	6
2	17	4	8	16	34	68
$\Sigma x=3$	$\Sigma y=24$	$\Sigma x^2=5$	$\Sigma x^3=9$	$\Sigma x^4=17$	$\Sigma xy=40$	$\Sigma x^2*y=74$

a0=1.00
a1=2.00
a2=3.00

Discussion:

Here I have discussed the Least Square Curve Fitting Problem where I have mentioned Linear Curve like Straight-line(2D) and Non-linear curve like polynomial where the equation is a parabolic function (The highest power of x is 2).